# Flutter - Introduction to Dart Programming

Dart is an open-source general-purpose programming language. It is originally developed by Google. Dart is an object-oriented language with C-style syntax. It supports programming concepts like interfaces, classes, unlike other programming languages Dart doesnt support arrays. Dart collections can be used to replicate data structures such as arrays, generics, and optional typing.

The following code shows a simple Dart program −

```
void main() {
   print("Dart language is easy to learn");
}
```

## Variables and Data types

Variable is named storage location and Data types simply refers to the type and size of data associated with variables and functions.

Dart uses var keyword to declare the variable. The syntax of var is defined below,

```
var name = 'Dart';
```

The final and const keyword are used to declare constants. They are defined as below −

```
void main() {
   final a = 12;
   const pi = 3.14;
   print(a);
   print(pi);
}
```

Dart language supports the following data types −

- **Numbers** − It is used to represent numeric literals Integer and Double.
- **Strings** − It represents a sequence of characters. String values are specified in either single or double quotes.
- **Booleans** − Dart uses the bool keyword to represent Boolean values true and false.
- **Lists and Maps** − It is used to represent a collection of objects. A simple List can be defined as below −.

```
void main() {
   var list = [1,2,3,4,5];
   print(list);
}
```

The list shown above produces [1,2,3,4,5] list.

Map can be defined as shown here −

```
void main() {
   var mapping = {'id': 1,'name':'Dart'};
   print(mapping);
}
```

- **Dynamic** − If the variable type is not defined, then its default type is dynamic. The following example illustrates the dynamic type variable −

```
void main() {
   dynamic name = "Dart";
   print(name);
}
```

## Decision Making and Loops

A decision making block evaluates a condition before the instructions are executed. Dart supports If, If..else and switch statements.

Loops are used to repeat a block of code until a specific condition is met. Dart supports for, for..in , while and do..while loops.

Let us understand a simple example about the usage of control statements and loops −

```
void main() {
   for( var i = 1 ; i <= 10; i++ ) {
      if(i%2==0) {
         print(i);
      }
   }
}
```

The above code prints the even numbers from 1 to 10.

## Functions

A function is a group of statements that together performs a specific task. Let us look into a simple function in Dart as shown here −

```
void main() {
   add(3,4);
}
void add(int a,int b) {
   int c;
   c = a+b;
   print(c);
}
```

The above function adds two values and produces 7 as the output.

## Object Oriented Programming

Dart is an object-oriented language. It supports object-oriented programming features like classes, interfaces, etc.

A class is a blueprint for creating objects. A class definition includes the following −

- Fields
- Getters and setters
- Constructors
- Functions

Now, let us create a simple class using the above definitions −

```
class Employee {
   String name;

   //getter method
   String get emp_name {
      return name;
   }
   //setter method
   void set emp_name(String name) {
      this.name = name;
   }
   //function definition
   void result() {
      print(name);
   }
}
void main() {
   //object creation
   Employee emp = new Employee();
   emp.name = "employee1";
   emp.result(); //function call
```



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